MACHINE LEARNING ENGINEER PATH

Python & Statistics to ML/DL Career Roadmap

Customized step-by-step roadmap leveraging your NumPy, Pandas, Matplotlib, Seaborn & Sheryians AI School Statistics foundations.

Your Starting Point (Foundations Completed):

1. Data Stack	2. Data Visualization	3. Mathematics & Stats
Python, NumPy, Pandas	Matplotlib, Seaborn	Probability, Hypothesis, Stats

PHASE 1: CLASSICAL MACHINE LEARNING (WEEKS 1–6)

Transition statistics into algorithms (Regression, Classification, Tree Models) using Scikit-Learn.

Step	Resource / Digital Book	Learning Objective & Connection	Free Link
1.1	**An Introduction to Statistical Learning (ISLP - Python)**	Connects probability and hypothesis testing directly to Supervised ML algorithms (Linear/Logistic Regression, Decision Trees).	Official PDF
1.2	**Hands-On Machine Learning (GitHub Companion)**	Use Pandas/NumPy data pipelines to train models with Scikit-Learn via interactive Colab notebooks.	GitHub Colab
1.3	**Scikit-Learn Official Interactive MOOC**	Master data preprocessing, feature scaling, model selection, and K-Fold cross-validation built by core maintainers.	Interactive Course

PHASE 2: DEEP LEARNING & NEURAL NETWORKS (WEEKS 7–12)

Build and train neural network architectures for Computer Vision and Natural Language Processing.

Step	Resource / Digital Book	Learning Objective & Connection	Free Link
2.1	**Dive into Deep Learning (d2l.ai)**	Interactive code-first book. Explains computational graphs, backpropagation, CNNs, and RNNs in PyTorch/NumPy.	Interactive Web
2.2	**Deep Learning with Python (Web Edition)**	Written by Keras creator François Chollet. Focuses on intuitive deep learning without complex mathematical noise.	Web Edition
2.3	**Deep Learning (The "DL Bible")**	By Ian Goodfellow et al. Use strictly as a reference text for loss functions, backprop math, and optimization theory.	Free Web Book

PHASE 3: SOFTWARE ENGINEERING & MODEL DEPLOYMENT (WEEKS 13–16)

Step	Resource / Digital Book	Learning Objective & Connection	Free Link
3.1	**Machine Learning Yearning (Andrew Ng)**	Structuring ML projects, choosing train/dev/test splits, analyzing bias vs. variance, and debugging model performance.	Official PDF
3.2	**FastAPI & Streamlit Deployment**	Package trained models into REST APIs (FastAPI) and host interactive web applications on Hugging Face Spaces.	FastAPI Docs

Will completing this make you a Job-Ready ML Engineer?
Yes. With your existing Python and Statistics background, completing this roadmap, building 3–4 Kaggle projects, and deploying APIs on GitHub makes you 100% qualified for Junior/Entry-Level ML Engineer roles.